AI in Automotive

1.Physical AI that act on External signal

Use case –Advanced Drier assistant system

Sense: ADAS captures information about the world in real time and accurately perceives and monitors the surrounding environment. That includes the road's physical characteristics and lane markings, and the vehicles nearby. It must distinguish between a small car, a truck, a bike and people.

Think: Computations use the system's perceptions to make intelligent decisions. With all of those inputs, in addition to data on weather conditions and other external factors, an ADAS must calculate fast — ideally, faster than a human can.

Act: Intelligence becomes action. The ADAS needs automation and optimization. It must respond in order to actuate a vehicle, slam on the brakes or change lanes, or do whatever is required

In energy industries, physical AI can synthesize weather data, pipeline pressure readings and other critical data in real time, providing operators with contextual awareness and predictive alerts about vulnerable system components

radar with artificial intelligence (AI) and machine learning (ML),

Today, Aptive has taken automotive radar even further. By enhancing radar with artificial intelligence (AI) and machine learning (ML), Aptiv has developed a groundbreaking radar-based object-classification system that achieves five times better performance on a broad set of radar sensors.